



**SCHOOL OF ELECTRICAL ENGINEERING
AND TELECOMMUNICATIONS**

**Resource Allocation for Integrated
Sensing and Communication (ISAC)
in Cell-Free Massive MIMO Systems**

by

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Thesis submitted as a requirement for the degree
Master of Engineering (Telecommunications)

Submitted: August 3, 2026

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Abstract

This thesis studies robust power-minimizing resource allocation for a cell-free massive MIMO integrated sensing and communication network. Distributed access points cooperatively serve communication users and transmit target-specific sensing waveforms under norm-bounded channel-state uncertainty. The proposed formulation jointly guarantees worst-case communication SINR, sensing SINR, and a matrix PCRB constraint while selecting a fixed-size sensing cluster for each active target. In contrast to formulations that credit all transmit covariance to sensing, the tracking FIM here is generated only by the known waveform dedicated to the corresponding target. The resulting mixed-integer non-convex problem is transformed through covariance lifting, an S-Procedure LMI, an exact Schur-complement PCRB LMI, and joint rank-one/binary DC penalties. A topology-aware SCA recovery procedure finally re-solves fixed binary candidates and validates the extracted physical beamformers.

Acknowledgements

I wish to thank all the people who have made contributions to the contents of this document; in particular Dr. Shaokang Hu and Prof. Derrick Wing Kwan Ng.

Abbreviations

6G Sixth-Generation wireless networks

AP Access Point

AoD Angle of Departure

AWGN Additive White Gaussian Noise

B5G Beyond 5th Generation wireless networks

CF-mMIMO Cell-Free Massive MIMO

CF-mMIMO Cell-Free Massive Multiple-Input Multiple-Output

CoMP Coordinated Multi-Point

CPU Central Processing Unit

CRB Cramér-Rao Bound

CSI Channel State Information

CU Communication User

CVLM Constant Velocity Linear Motion

CVX Convex Optimization (software package)

DL Downlink

DPI Direct-Path Interference

EE&T School of Electrical Engineering and Telecommunications

FD Full-Duplex

FIM Fisher Information Matrix

IoT Internet of Things

ISAC Integrated Sensing and Communication

JCAS Joint Communication and Sensing

LoS Line-of-Sight

LP Linear Programming

MATLAB Matrix Laboratory (programming environment)

ME Master of Engineering

MIMO Multiple-Input Multiple-Output

MINLP Mixed-Integer Non-Linear Programming

mmWave Millimeter Wave

NLoS Non-Line-of-Sight

NP-hard Non-deterministic Polynomial-time hard

PCRB Posterior Cramér-Rao Bound

PDF Probability Density Function

PoD Probability of Detection

QoS Quality of Service

RCS Radar Cross-Section

RF Radio Frequency

RF Radio Frequency

SCA Successive Convex Approximation

SDR Semidefinite Relaxation

SDP Semidefinite Programming

SBS Single Base Station

SIC Self-Interference Cancellation

SINR Signal-to-Interference-plus-Noise Ratio

SLA Service Level Agreement

SLAM Simultaneous Localization and Mapping

THz Terahertz

UAV Unmanned Aerial Vehicle

UAV Unmanned Aerial Vehicle

UL Uplink

V2X Vehicle-to-Everything

Notation

Bold lower-case and upper-case letters denote vectors and matrices, respectively. $(\cdot)^T$, $(\cdot)^H$, $\text{tr}(\cdot)$, and $\|\cdot\|_2$ denote transpose, Hermitian transpose, trace, and the Euclidean norm.

$\mathcal{M}, \mathcal{K}, \mathcal{P}$ Sets of APs, communication users, and sensing targets.

M, K, P Numbers of APs, communication users, and sensing targets.

$N_t, N = MN_t$ Antennas per AP and total stacked transmit dimension.

$\mathcal{P}^{\text{active}}$ Targets requiring sensing service in the current scheduling interval.

N_{req} Number of APs authorized for each active target.

$\mathbf{w}_{m,k}, \mathbf{w}_k$ Beamformer from AP m to user k , and its network-wide stacked form.

$\mathbf{W}_k$ Lifted communication covariance, $\mathbf{w}_k \mathbf{w}_k^H$.

$\mathbf{S}_p$ Network-wide covariance of the dedicated sensing waveform for target p .

$\mathbf{R}_X$ Total transmit covariance, $\sum_k \mathbf{W}_k + \sum_p \mathbf{S}_p$.

b_{mp} Binary authorization indicator for AP m to allocate dedicated sensing power to target p .

$\mathbf{E}_m$ Diagonal matrix selecting the antenna block of AP m .

$\hat{\mathbf{h}}_k, \Delta \mathbf{h}_k, \mathbf{h}_k$ Nominal channel, channel error, and uncertain stacked channel of user k .

ϵ_h Relative radius of the norm-bounded CSI uncertainty set.

$\mathbf{g}_p, \mathbf{D}_p$ Sensing response vector and its derivative matrix for target p .

$\mathbf{J}_p^{\text{data}}, \mathbf{M}_p$ Data FIM and auxiliary matrix used in the PCRB Schur-complement LMI.

$\gamma_k, \gamma_S^{\text{PoD}}$ Required communication SINR and sensing-SINR threshold.

$\Gamma_{\text{Track},p}$ Maximum permitted trace PCRB for target p .

σ_c^2, σ_s^2 Communication-noise and sensing-noise powers.

P_{max} Per-AP total transmit-power budget.

μ_k Nonnegative S-Procedure multiplier for user k .

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Chapter 1

Introduction

1.1 Motivation

Sixth-generation wireless networks must provide reliable connectivity and timely awareness of the surrounding environment for applications such as industrial automation, intelligent transport, and autonomous aerial systems (Dong et al., 2023; Kaushik et al., 2024). (F. Liu, Cui, et al., 2022; F. Liu et al., 2020) Integrated sensing and communication (ISAC) addresses this requirement by sharing spectrum, radio hardware, and spatial degrees of freedom between data delivery and sensing. The resulting benefit is compelling, but the shared resources also create a fundamental allocation problem: power assigned to communication, dedicated sensing, and interference protection cannot be chosen independently.

Cell-free massive MIMO makes this problem both more promising and more challenging. A central processing unit can coordinate distributed access points (APs) to serve users cooperatively, thereby creating spatial diversity for communication and sensing without fixed cell boundaries (Demirhan & Alkhateeb, 2025; Femenias & Riera-Palou, 2025). (Li et al., 2024; Mao et al., 2024) However, each AP has a finite transmit-power budget. Selecting which APs may illuminate each target therefore changes the sensing geometry and the remaining power available for globally cooperative communication. The decision must also remain reliable when the communication channel state information (CSI) is imperfect.

1.2 Research Gap, Problem, and Contributions

Existing cooperative ISAC formulations establish the value of spatial cooperation, yet three connected gaps remain for the design considered here. First, a sensing association is often difficult to interpret when the model does not distinguish a target’s dedicated sensing waveform from the globally cooperative communication signal. Second, perfect-CSI formulations can lose their communication quality-of-service (QoS) guarantee under channel error. Third, a relaxed semidefinite or continuous association solution is not itself a physical beamforming and topology design; rank-one and binary recovery must be assessed after the relaxation is solved.

This thesis addresses these gaps through robust downlink resource allocation for a cell-free ISAC network. The objective is to minimize total transmit power while satisfying worst-case communication SINR, target detection, and tracking accuracy requirements. The central design choice is a known, target-specific sensing covariance for each target. Its binary AP–target association variable authorizes dedicated sensing power only; it does not gate the globally cooperative communication beamformers. This separation makes the topology decision physically interpretable and permits each target’s Fisher information and PCRB requirement to be evaluated from its own probing covariance.

1.3 Thesis Organization

Chapter 2 narrows the literature from cooperative ISAC to the four design requirements that motivate this thesis: cooperative resource allocation, robust QoS, multi-target tracking accuracy, and recoverable sensing topology. Chapter 3 then defines the network roles, the robust communication model, and the target-specific sensing model. Chapter 4 formulates the power-minimization problem and derives the lifted LMI and dual-DC recovery framework. Chapter 5 evaluates the proposed workflow through topology, robustness, recovery, and scalability evidence. Finally, Chapter 6 answers the research problem from the accumulated evidence and identifies the validation work that remains.

Chapter 2

Literature Review

This chapter positions the thesis around the decisions that must be made by a robust cell-free ISAC transmitter. Rather than treating ISAC architectures, sensing metrics, and optimization methods as separate topics, the review follows the same chain as the proposed design: cooperation creates useful spatial degrees of freedom, QoS protection constrains their use, target-state accuracy determines the sensing requirement, and topology recovery determines whether the computed solution is physically implementable.

2.1 From Coexistence to Cooperative ISAC

ISAC emerged from the observation that wireless communication and sensing rely on the same electromagnetic propagation, spectrum, and radio hardware. Early single-base-station systems therefore studied waveform sharing, communication–sensing trade-offs, and self-interference management (He et al., 2023; Hua et al., 2023; A. Liu et al., 2022). (F. Liu, Cui, et al., 2022; F. Liu et al., 2020) These studies establish the core trade-off, but a single transmitter provides limited geometric diversity and can be vulnerable to cell-edge blockage.

Cooperative and multistatic ISAC extend the sensing and communication aperture across multiple base stations or APs. Spatially distributed observations can improve coverage, localization geometry, and interference management, provided that the transmitters are coordinated (Cui et al., 2025; Liao et al., 2025; Meng et al., 2025). (Cheng et al., 2024; Li et al., 2024; Mao et al., 2024) Cell-free massive MIMO takes this idea further: a large set of distributed APs is centrally coordinated without fixed cell boundaries (Demirhan & Alkhateeb, 2025; Femenias & Riera-Palou, 2025). (Mao et al., 2024; Ren et al., 2024) The same cooperation that improves

service flexibility also couples every AP’s communication and sensing power decisions. This is the architectural setting of the present thesis.

2.2 Design Requirements for Robust Multi-Target Sensing

2.2.1 Communication QoS under Imperfect CSI

Communication QoS is commonly expressed through rate or SINR constraints. Many ISAC formulations assume perfect CSI to simplify joint beamforming, but estimation error, feedback delay, and quantization make this assumption fragile in practice (Chu et al., 2023; Yu et al., 2022). (Gong et al., 2025; He et al., 2024) For a cell-free transmitter, channel uncertainty is particularly consequential: the dedicated sensing covariances and multi-user communication beams share the same AP power budgets and jointly determine the user interference level. This motivates a norm-bounded worst-case SINR requirement rather than a nominal-channel guarantee.

2.2.2 Target-Specific Detection and Tracking Accuracy

Sensing designs are often evaluated with sensing SINR and Cramér–Rao-bound (CRB) metrics. Sensing SINR is a detection-related proxy, whereas the CRB measures a lower bound on estimation error and is consequently useful for localization and tracking-oriented design (Geng et al., 2025; Liao et al., 2025). (F. Liu, Liu, et al., 2022) For multiple targets, however, the sensing resource assigned to one target should not be credited to another merely because the network uses a shared transmit infrastructure. The present model therefore assigns a known sensing covariance to each target and forms its FIM and matrix PCRB from that covariance alone. This choice makes the association and tracking requirements refer to the same physical sensing resource.

2.2.3 Sensing Topology and Physical Recovery

Cell-free systems can authorize subsets of APs to allocate a sensing waveform for a target, trading sensing geometry against finite AP power and implementation cost. This turns sensing-topology authorization into a mixed-integer design variable rather than a post-processing detail. Semidefinite relaxation is valuable for handling beamforming structure, but a relaxed covari-

ance or a continuous association weight does not by itself establish an implementable design. A complete workflow must recover rank-one beamformers and binary associations, then test the recovered variables against the original communication, sensing, tracking, and power constraints.

Recent work also broadens ISAC across waveform, hardware, and surface-assisted architectures: OTFS exploits delay–Doppler structure for high-mobility sensing and communication (Yuan et al., 2021); one-bit transceiver design addresses low-resolution mmWave hardware (Lin et al., 2025); and STAR-RIS and holographic surfaces provide relevant near-field extensions (Wang et al., 2025; Zhang et al., 2022). At the system layer, sensing-aware status updating offers a complementary scheduling view (Peng et al., 2023). These directions motivate the scope of this thesis but are not assumed by its conventional multi-antenna AP signal model.

2.3 Research Gap and Thesis Position

The preceding literature motivates a precise gap at the intersection of the four requirements. Cooperative ISAC provides the relevant spatial degrees of freedom; robust QoS prevents their allocation from relying on nominal CSI; target-specific FIM/PCRB modelling makes dedicated sensing participation meaningful; and recovery-aware optimization separates a feasible relaxed point from a validated physical design. This thesis combines these elements in one power-minimizing cell-free ISAC formulation. Chapter 3 defines the required resource separation, and Chapter 4 develops the corresponding robust and recovery-aware optimization method.

Chapter 3

System Model

The model is organized around a deliberate resource separation. Communication is a network-wide cooperative service, whereas sensing is accounted for through a distinct, known covariance for each target. The AP–target association will therefore answer a narrow physical question—which APs may allocate dedicated sensing power to a target—without artificially restricting the communication beamforming support. The following sections introduce this separation in the same order in which it enters the optimization problem: network resources, robust communication protection, and target-specific sensing accuracy.

3.1 Network and Signalling Model

We consider a downlink cell-free integrated sensing and communication (ISAC) network in which a central processing unit coordinates M distributed access points (APs). Each AP has N_t transmit antennas and jointly serves K single-antenna communication users while illuminating P sensing targets. The AP, user, and target index sets are denoted by $\mathcal{M} = \{1, \dots, M\}$, $\mathcal{K} = \{1, \dots, K\}$, and $\mathcal{P} = \{1, \dots, P\}$, respectively. A subset $\mathcal{P}^{\text{active}} \subseteq \mathcal{P}$ contains the targets that require sensing service in the current scheduling interval. The model is instantaneous within one coherence block; target-state prediction across blocks is outside the scope of the resource-allocation problem.

Let $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$ be the communication beamformer from AP m to user k . To make sensing participation explicit, a distinct zero-mean sensing waveform is assigned to every target. Its network-wide stacked covariance is $\mathbf{S}_p \succeq \mathbf{0}$, where the stacked transmit dimension is $N =$

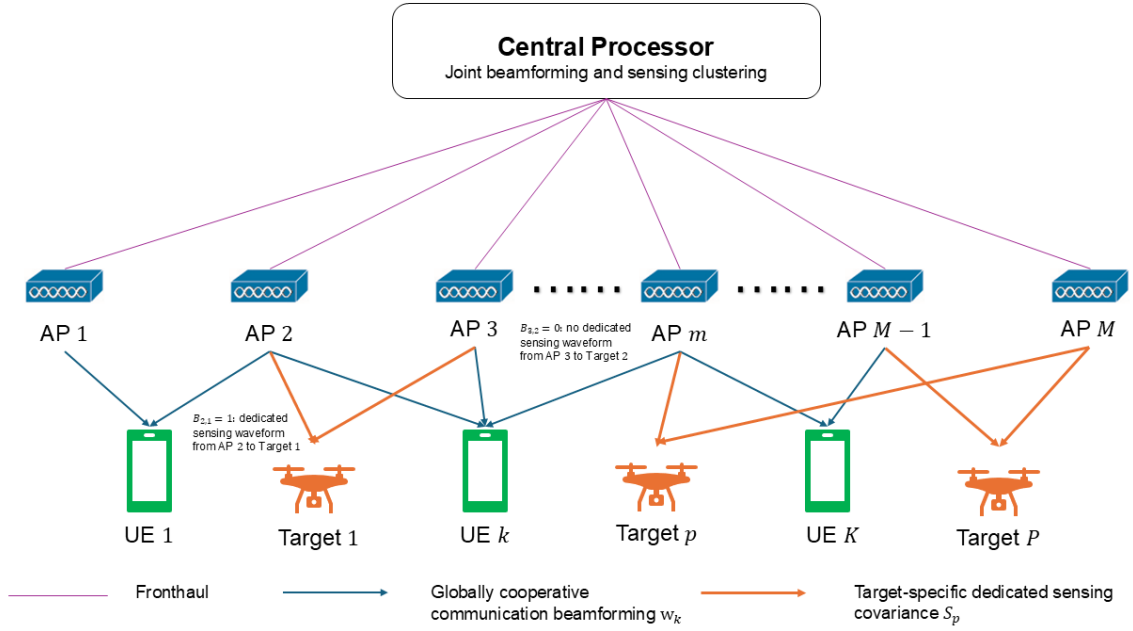


Figure 3.1: Cell-free ISAC architecture. The CPU coordinates distributed APs through fronthaul/control links (magenta). APs jointly serve users through globally cooperative communication beamforming (blue), whereas only associations with $b_{mp} = 1$ authorize the target-specific dedicated sensing covariance S_p (orange). The generic-index illustration illustrates an inactive association, $b_{3,2} = 0$, which denies AP 3 dedicated sensing power for target 2 only; it does not prevent AP 3 from contributing to communication.

MN_t . Define the stacked communication beamformer

$$\mathbf{w}_k \triangleq [\mathbf{w}_{1,k}^T, \dots, \mathbf{w}_{M,k}^T]^T \in \mathbb{C}^N, \quad (3.1)$$

and its covariance $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$. The total transmit covariance is therefore

$$\mathbf{R}_X = \sum_{k \in \mathcal{K}} \mathbf{W}_k + \sum_{p \in \mathcal{P}} \mathbf{S}_p. \quad (3.2)$$

Let $\mathbf{E}_m \in \mathbb{R}^{N \times N}$ select the N_t -antenna block of AP m . The per-AP hardware constraint is

$$\text{Tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}. \quad (3.3)$$

3.2 Robust Communication Model

The nominal stacked downlink channel of user k is $\hat{\mathbf{h}}_k \in \mathbb{C}^N$. Imperfect channel state information (CSI) is represented by the norm-bounded uncertainty model

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta \mathbf{h}_k, \quad \|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2. \quad (3.4)$$

Let $\mathbf{S}_\Sigma \triangleq \sum_{p \in \mathcal{P}} \mathbf{S}_p$ denote the aggregate dedicated sensing covariance. It is treated as interference at a communication user. This conservative and implementable assumption avoids requiring receiver-side cancellation of the sensing waveforms. Thus the worst-case downlink SINR requirement is

$$\min_{\|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2} \frac{|(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_k|^2}{\sum_{j \neq k} |(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_j|^2 + (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{S}_\Sigma (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k) + \sigma_c^2} \geq \gamma_k, \quad \forall k \in \mathcal{K}, \quad (3.5)$$

where σ_c^2 is the communication-noise power and γ_k is the required SINR of user k .

3.3 Target-Specific Sensing Model

The binary variable $b_{mp} \in \{0, 1\}$ indicates whether AP m is admitted to the sensing cluster of target p . It authorizes dedicated sensing power only; it does *not* restrict the globally cooperative communication beamformers. For each active target, exactly N_{req} APs are admitted:

$$\sum_{m \in \mathcal{M}} b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}}. \quad (3.6)$$

The target-specific association gate is

$$\text{Tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. \quad (3.7)$$

Consequently, an unselected AP cannot allocate sensing power to target p , whereas a selected AP is not forced to consume a minimum amount of power. In particular, (3.7) is not a gate on communication power and must be read jointly with the total hardware limit (3.3).

Let $\mathbf{g}_p \in \mathbb{C}^N$ be the effective sensing steering channel of target p . The target-specific sensing SINR is

$$\text{SINR}_{S,p} = \frac{\mathbf{g}_p^H \mathbf{S}_p \mathbf{g}_p}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}, \quad \forall p \in \mathcal{P}, \quad (3.8)$$

where σ_s^2 is the sensing-noise power and γ_S^{PoD} is the detection-related threshold.

For a target state $\boldsymbol{\theta}_p \in \mathbb{R}^{N_\theta}$, let $\mathbf{D}_p = \partial \mathbf{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{N \times N_\theta}$ be the effective response derivative. Under the equivalent MISO sensing observation model, the data Fisher information matrix (FIM) is

$$\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\}. \quad (3.9)$$

Only the known waveform covariance $\mathbf{S}_p$ receives sensing-information credit. Communication covariances and waveforms dedicated to other targets are not counted in (3.9). Tracking accuracy is constrained by the matrix PCRB/CRB requirement

$$\text{Tr}((\mathbf{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}. \quad (3.10)$$

For $N_\theta = 1$, this reduces to the scalar lower bound $J_p^{\text{data}} \geq 1/\Gamma_{\text{Track},p}$; for multi-dimensional states, the full matrix form is retained. The equivalent semidefinite form used to solve (3.10) is developed in Chapter 4.

Design implication. The model couples the three services through the common AP-power budget but keeps their roles distinguishable: communication is protected for every admissible CSI error, while a target receives sensing credit only from its own authorized covariance. Chapter 4 turns these requirements into the power-minimization problem and identifies the robust, matrix, rank, and binary structures that prevent a direct convex solution.

Chapter 4

Optimization Formulation and Solution

Chapter 3 identifies the physical allocation question: minimize total transmit power while preserving globally cooperative communication, target-specific sensing quality, and AP-level implementability. This chapter keeps that interpretation visible throughout the transformation. Problem (P1) states the original binary physical design; (P2) lifts the beamforming variables and relaxes only the binary box for optimization; and (P3) collects the robust and tracking LMIs with the DC penalties used for candidate recovery. Thus, an LMI-feasible point is an intermediate optimization object, while a recovered design is accepted only after the physical checks used in Chapter 5.

4.1 Problem Formulation

Target-Centric Dedicated Sensing Associations. The binary variable $b_{mp} \in \{0, 1\}$ is a sensing-cluster admission indicator: it specifies whether AP m is authorized to allocate dedicated sensing resources to target p . Let $\mathbf{S}_p \succeq \mathbf{0}$ denote the covariance of the corresponding waveform. When $b_{mp} = 1$, AP m may allocate transmit power to $\mathbf{S}_p$; when $b_{mp} = 0$, its corresponding covariance block is forced to zero. Thus, $b_{mp} = 1$ is an authorization rather than a minimum-power guarantee, and realized participation is quantified from the nonzero sensing-power blocks after optimization. The indicator does not restrict communication beamforming, which remains globally cooperative. The per-AP transmit-hardware constraint is

$$\text{Tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}. \quad (4.1)$$

The association constraint and the dedicated sensing resource constraint are

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \text{Tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\text{max}} b_{mp}, \quad \forall m, p. \quad (4.2)$$

The cardinality constraint specifies a fixed sensing-cluster size for each target. If N_{req} is intended as an upper bound rather than a fixed budget, the equality should be replaced by $\leq N_{\text{req}}$ and accompanied by an explicit fronthaul or processing cost.

The total transmit covariance is $\mathbf{R}_X = \sum_{k=1}^K \mathbf{W}_k + \sum_{p=1}^P \mathbf{S}_p$. To retain a known and target-specific probing signal at the sensing receiver, the PCRB of target p is evaluated only from $\mathbf{S}_p$:

$$\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\}. \quad (4.3)$$

Thus, the association variable controls the dedicated sensing resources that can contribute to the PCRB, whereas communication covariances do not receive PCRB credit. The constraints above are affine in the continuous covariance variables for fixed b_{mp} and linear in b_{mp} for fixed covariances. Unless receiver-side cancellation is explicitly assumed, the aggregate dedicated sensing covariance $\sum_p \mathbf{S}_p$ is included as interference in the communication SINR. The Sum Big-M power-gating formulation and derivations based on an FIM that is affine solely in $\mathbf{R}_X$ correspond to a different model and should be replaced by the present formulation.

The decision variables of the original problem (P1) are the per-AP beamforming vectors $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$, the target-specific sensing covariance matrices $\mathbf{S}_p \in \mathbb{H}_+^{MN_t}$, and the binary AP–

target association indicators $b_{mp} \in \{0, 1\}$:

$$\min_{\substack{\{\mathbf{w}_{m,k}\}_{m \in \mathcal{M}, k \in \mathcal{K}}, \{\mathbf{S}_p\}_{p \in \mathcal{P}}, \\ \{b_{mp}\}_{m \in \mathcal{M}, p \in \mathcal{P}}}} \sum_{m=1}^M \sum_{k=1}^K \|\mathbf{w}_{m,k}\|^2 + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p) \quad (\text{P1-objective})$$

$$\text{s.t.} \quad \min_{\|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|} \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \mathbf{h}_k^H \left(\sum_p \mathbf{S}_p \right) \mathbf{h}_k + \sigma_c^2} \geq \gamma_k, \quad \forall k, \quad (\text{P1-C1})$$

$$\text{SINR}_{S,p} := \frac{\mathbf{g}_p^H \mathbf{S}_p \mathbf{g}_p}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}, \quad \forall p \in \mathcal{P}, \quad (\text{P1-C2})$$

$$\text{Tr}((\mathbf{J}_p^{\text{data}}(\mathbf{S}_p))^{-1}) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}, \quad (\text{P1-C3})$$

$$\text{Tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}, \quad (\text{P1-C4'a})$$

$$\text{Tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}, \quad (\text{P1-C4'b})$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}}, \quad (\text{P1-C5'})$$

$$\mathbf{S}_p \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (\text{P1-C6})$$

$$b_{mp} \in \{0, 1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. \quad (\text{P1-C7'})$$

Here $\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta \mathbf{h}_k$ is the uncertain stacked channel from (3.4). The cooperative stacked beamforming vector is $\mathbf{w}_k \triangleq [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \dots, \mathbf{w}_{M,k}^T]^T \in \mathbb{C}^{MN_t}$, obtained by stacking the per-AP variables. Further, $\mathbf{R}_X \triangleq \sum_{k=1}^K \mathbf{w}_k \mathbf{w}_k^H + \sum_{p=1}^P \mathbf{S}_p$ (precisely defined in the lifted form of §4.2-A), and $\mathbf{E}_m \in \mathbb{R}^{MN_t \times MN_t}$ is the diagonal selection matrix that extracts the antenna block of AP m from $\mathbf{R}_X$. The FIM $\mathbf{J}_p^{\text{data}}$ in (P1-C3) takes the following original form for a multi-dimensional target state $\boldsymbol{\theta}_p \in \mathbb{R}^{N_\theta}$ (typical ISAC scenarios involve 2D/3D position, or angle+range with $N_\theta \in \{2, 3, 4\}$):

$$\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\} \in \mathbb{H}_+^{N_\theta \times N_\theta}, \quad (4.4)$$

where $\mathbf{D}_p \triangleq \partial \mathbf{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{MN_t \times N_\theta}$ is the effective target-response derivative matrix. The equivalent MISO observation model uses the dedicated sensing covariance directly; hence (4.4) restates (4.3) and is affine in $\mathbf{S}_p$ only—the communication covariances $\{\mathbf{W}_k\}$ receive no PCRB credit. The CRB matrix is $\mathbf{C}_p = (\mathbf{J}_p^{\text{data}})^{-1}$ and (P1-C3) requires $\text{Tr}(\mathbf{C}_p) \leq \Gamma_{\text{Track},p}$. This matrix trace-of-inverse constraint must be retained for $N_\theta \geq 2$; for $N_\theta = 1$, it reduces to the scalar condition $\mathbf{J}_p^{\text{data}} \geq 1/\Gamma$. The association gate (P1-C4'a) and the hardware ceiling (P1-C4'b) are both linear: (P1-C4'a) authorizes AP m to allocate dedicated sensing power to target p up to $P_{\max}$ when $b_{mp} = 1$ and forces the corresponding block of $\mathbf{S}_p$ to zero when $b_{mp} = 0$, without

gating communication transmission, while (P1-C4'b) caps the total per-AP transmit covariance $\mathbf{R}_X$.

Problem (P1) is non-convex. Its non-convexity stems from four independent sources distributed across different constraints. The communication worst-case SINR constraint (P1-C1) carries a fractional quadratic structure: the numerator $|\mathbf{h}_k^H \mathbf{w}_k|^2$ and the denominator $\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \mathbf{h}_k^H (\sum_p \mathbf{S}_p) \mathbf{h}_k + \sigma_c^2$ are both quadratic functions of the beamforming vectors and the sensing covariances, so their ratio is a non-convex function on the manifold where the denominator is positive [1]. Compounding this, the worst-case quantifier $\forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$ is semi-infinite, and the resulting set $\{\mathbf{w} : \min_{\Delta \mathbf{h} \in \mathcal{B}_\epsilon} \text{SINR}_k(\Delta \mathbf{h}) \geq \gamma_k\}$ is not closed under convex combinations. The position PCRB constraint (P1-C3) requires $\text{Tr}((\mathbf{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track},p}$; although the composite function $\text{Tr}(\mathbf{J}^{-1})$ is strictly convex on $\mathbf{J} \succ \mathbf{0}$, so that (P1-C3) itself defines a convex set, the constraint involves the nonlinear matrix inverse and must be transformed into an LMI via the Schur complement before standard SDP solvers can handle it. The binary AP–target association indicators b_{mp} in (P1-C7') can be characterized on the relaxed box by $g(b) = b - b^2 = 0$; this characterization is used later as a concave penalty since $g(b) \geq 0$ for $b \in [0, 1]$ and g is strictly concave. The association gate (P1-C4'a) and the per-AP hardware ceiling (P1-C4'b) are both linear: the left-hand sides are the linear trace terms $\text{Tr}(\mathbf{E}_m \mathbf{S}_p)$ and $\text{Tr}(\mathbf{E}_m \mathbf{R}_X)$ extracted by the diagonal selection matrix $\mathbf{E}_m$, and the right-hand sides are $P_{\max} b_{mp}$ (association-coupled) and $P_{\max}$ (standard hardware upper bound) respectively; both constraints are mixed-integer affine in (P1) and become jointly affine after the box relaxation $0 \leq b_{mp} \leq 1$ used in (P2). The PSD cone $\mathbf{S}_p \succeq \mathbf{0}$ itself ((P1-C6)), the linear sensing SINR $\text{Tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p) \geq \gamma_S^{\text{PoD}} \sigma_s^2$ ((P1-C2)), and the binary association constraint (P1-C7') is discrete, but because of the four non-convex/nonlinear sources above the overall problem remains a non-convex MINLP. In §4.2 we address each source by a dedicated step of the convexification chain (the rank-1 constraint that arises after covariance lifting is treated as a fifth source and is recovered jointly with the binarity of b_{mp} by the dual DC penalty SCA framework of §3; the binary condition belongs to (P1), while its box relaxation and DC penalty are introduced only after lifting in §4.2-E).

4.2 Solution of the Optimization Problem

This section solves the problem along the five non-convex/nonlinear sources identified at the end of §4.1. The SINR fraction and the semi-infinite uncertainty are converted into LMIs via

cross-multiplication (§4.2-C step 1) followed by the S-Procedure (§4.2-C step 2); the trace of the PCRB matrix inverse is losslessly converted into an LMI by the Schur complement (§4.2-D); the binarity of the AP–target soft-assignment weights b_{mp} is handled by continuous relaxation plus a DC penalty added to the objective (§4.2-E), which shares a unified dual-DC-penalty SCA framework with the rank-1 constraint in §3.

A. Problem Transformation (P2)

To prepare for the semidefinite relaxation, we lift the vector variables in (P1) into covariance matrices, i.e. we define $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H \in \mathbb{H}_+^{MN_t}$ and $\mathbf{R}_X \triangleq \sum_{k=1}^K \mathbf{W}_k + \sum_{p=1}^P \mathbf{S}_p$, where the target-specific sensing covariances $\mathbf{S}_p \in \mathbb{H}_+^{MN_t}$ were introduced in §4.1, yielding (P2):

$$\min_{\substack{\{\mathbf{W}_k\}_{k \in \mathcal{K}}, \{\mathbf{S}_p\}_{p \in \mathcal{P}}, \\ \{b_{mp}\}_{m \in \mathcal{M}, p \in \mathcal{P}}}} \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p) \quad (\text{P2})$$

$$\begin{aligned} \text{s.t.} \quad & \text{Tr}(\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \mathbf{W}_k) - \gamma_k \sum_{j \neq k} \text{Tr}(\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \mathbf{W}_j) \\ & - \gamma_k \text{Tr}(\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \sum_p \mathbf{S}_p) \geq \gamma_k \sigma_c^2, \quad \forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|, \quad \forall k, \end{aligned} \quad (\text{P2-C1})$$

$$\text{Tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C2})$$

$$\text{Tr}((\mathbf{J}_p^{\text{data}}(\mathbf{S}_p))^{-1}) \leq \Gamma_{\text{Track}, p}, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C3})$$

$$\text{Tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}, \quad (\text{P2-C4'a})$$

$$\text{Tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}, \quad (\text{P2-C4'b})$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}}, \quad (\text{P2-C5'})$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \in \mathcal{K}, \quad (\text{P2-C6})$$

$$\mathbf{S}_p \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C7})$$

$$\text{rank}(\mathbf{W}_k) = 1, \quad \forall k \in \mathcal{K}, \quad (\text{P2-C8})$$

$$0 \leq b_{mp} \leq 1, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. \quad (\text{P2-C9})$$

Here $\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\} \in \mathbb{H}_+^{N_\theta \times N_\theta}$ is the data-dependent Fisher Information Matrix under the equivalent MISO sensing observation model, where $\mathbf{D}_p = \partial \mathbf{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{MN_t \times N_\theta}$ is the effective target-response derivative; $\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \triangleq (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H$. This model uses the dedicated sensing covariance $\mathbf{S}_p$ directly and does not posit a vectorized multi-receiver

architecture. Note that each $\mathbf{S}_p$ in (P2) is a joint stacked covariance, so the dedicated waveform for target p may still be synthesized cooperatively across APs; the association gate (P2-C4'a) only authorizes which APs may allocate power to $\mathbf{S}_p$, while (P2-C2) sensing SINR and (P2-C3) PCRB are both evaluated from $\mathbf{S}_p$ alone.

(P2) retains the exact covariance lifting of the beamforming variables: when $\text{rank}(\mathbf{W}_k) = 1$, the mapping $\mathbf{w}_k \leftrightarrow \mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$ is recoverable and the quadratic terms are rewritten through identities such as $\text{Tr}(\mathbf{x} \mathbf{x}^H \mathbf{W}) = \mathbf{x}^H \mathbf{W} \mathbf{x}$. However, replacing the binary constraint (P1-C7') by the box constraint (P2-C9) makes (P2) a continuous relaxation of (P1), not an equivalent reformulation. The relaxation can provide a lower bound, while the DC penalty and the subsequent physical-recovery step are used to recover and validate a binary design.

Problem (P2) is the lifted continuous relaxation of (P1). The robust SINR fraction, semi-infinite uncertainty, and PCRB matrix inverse remain to be transformed. The binary association constraint is relaxed to the box (P2-C9) and is recovered later by the DC penalty and physical-validation procedure. The rank-one condition (P2-C8) preserves recoverability of the lifted beamforming covariance but is itself non-convex; SDR subsequently drops this condition and therefore gives a further lower-bound relaxation. The dedicated sensing gate (P2-C4'a), total AP-power limit (P2-C4'b), PSD constraints, and sensing-SINR constraint are affine or convex in the lifted continuous variables.

B. Semi-Definite Relaxation

The NC4 in (P2) is addressed by SDR: the rank-1 manifold $\mathcal{F}_{\text{rank-1}}$ is itself non-convex, and SDR relaxes (P2-C8) to $\mathbf{W}_k \succeq \mathbf{0}$ only, i.e.

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k, \quad (\text{S1.1})$$

The feasible region is enlarged from $\mathcal{F}_{\text{rank-1}}$ to the positive semidefinite cone $\mathcal{F}_{\text{SDR}}$, and the SDR solution provides a lower bound on the original problem, $P_{\text{SDR}}^* \leq P_{\text{P2}}^*$. For the joint structure of per-user SINR + cell-free cooperation + robust uncertainty, neither the multi-group multicasting tightness condition $G_k \leq 2$ [2] nor the robust MISO condition $N_t \leq 2$ [3] directly applies, and Gaussian randomization must be used to recover a feasible rank-1 solution when $\text{rank}(\mathbf{W}_k^*) > 1$. Specifically, $L \in [100, 1000]$ Gaussian random vectors are generated and the candidate that satisfies the constraints and is optimal in the objective is selected; however, this post-processing has no tight performance-loss upper bound.

C. S-Procedure

This subsection handles the residual semi-infinite robust uncertainty in NC1 in two steps: first the original SINR robust inequality is cross-multiplied into a quadratic form (an equivalent rewrite of (P2-C1), no longer a fraction), and then the S-Procedure reduces the universal quantifier over $\Delta \mathbf{h}_k$ to an LMI in the S-Procedure relaxation variable μ_k .

Step 1 – Cross-multiplication. Fix any worst-case $\Delta \mathbf{h}_k$ satisfying $\|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$. Because $\sigma_c^2 > 0$ guarantees that the denominator is strictly positive, the SINR inequality can be cross-multiplied directly into the quadratic form

$$\text{Tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) - \gamma_k \sum_{j \neq k} \text{Tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) - \gamma_k \text{Tr} \left(\hat{\mathbf{H}}_k \sum_p \mathbf{S}_p \right) \geq \gamma_k \sigma_c^2, \quad (\text{S2.1})$$

which is a strict equivalence rather than an approximation; strict positivity of the denominator guarantees that the implication holds in both directions. Since σ_c^2 contains no decision variable, the fraction can be eliminated without a first-order Taylor approximation, thus avoiding the accumulation of SCA iteration error.

Step 2 – S-Procedure. Restoring $\Delta \mathbf{h}_k$ as a variable and defining $\mathbf{A}_k \triangleq \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j - \sum_p \mathbf{S}_p$, (P2-C1) is rewritten as a quadratic inequality in $\Delta \mathbf{h}_k$:

$$\tilde{f}_1(\Delta \mathbf{h}) = (\hat{\mathbf{h}} + \Delta \mathbf{h})^H \mathbf{A}_k (\hat{\mathbf{h}} + \Delta \mathbf{h}) - \sigma_c^2 \geq 0, \quad \forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|,$$

and the uncertainty set is described by $\tilde{f}_2(\Delta \mathbf{h}) = \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 - \|\Delta \mathbf{h}_k\|^2 \geq 0$. For this single quadratic constraint on a norm ball, the universal quantifier admits, via the S-Procedure [1], a finite LMI equivalent condition:

Lemma 4.1. (S-Procedure). Let $\tilde{f}_m(\Delta \mathbf{h}) = (\Delta \mathbf{h})^H \mathbf{A}_m (\Delta \mathbf{h}) + 2 \text{Re}\{(\Delta \mathbf{h})^H \mathbf{b}_m\} + c_m$, $m \in \{1, 2\}$. Then the implication “ $\tilde{f}_1(\Delta \mathbf{h}) \geq 0 \Rightarrow \tilde{f}_2(\Delta \mathbf{h}) \geq 0$ ” holds if and only if there exists $\mu \geq 0$ such that

$$\begin{bmatrix} \mathbf{A}_2 - \mu \mathbf{A}_1 & \mathbf{b}_2 - \mu \mathbf{b}_1 \\ \mathbf{b}_2^H - \mu \mathbf{b}_1^H & c_2 - \mu c_1 \end{bmatrix} \succeq \mathbf{0},$$

For a single norm-ball antecedent satisfying Slater’s condition, this certificate is lossless. In the application below, the uncertainty function is the antecedent and the SINR quadratic is the consequent.

Applying Lemma 4.1 to the implication $\tilde{f}_2 \geq 0 \Rightarrow \tilde{f}_1 \geq 0$, where $\tilde{f}_2$ describes the norm-ball uncertainty set and $\tilde{f}_1$ is the required SINR quadratic, the worst-case robust form of (P2-C1)

can be written as the following LMI:

$$\exists \mu_k \geq 0 : \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 \end{bmatrix} \succeq \mathbf{0}. \quad (\text{S3.1})$$

The newly introduced $\mu_k \geq 0$ is the S-Procedure relaxation variable.

D. PCRB Constraint: Schur Complement LMI Transformation

The PCRB constraint (P1-C3) remains in the lifted domain as

$$\text{Tr}((\mathbf{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C3})$$

where $\mathbf{J}_p^{\text{data}} \succ \mathbf{0}$ is an affine function of $\mathbf{S}_p$. The matrix-inverse map $\mathbf{J} \mapsto \mathbf{J}^{-1}$ is strictly matrix-convex on the positive-definite cone, and the trace operator $\text{Tr}(\cdot)$ is a linear order-preserving map; therefore the composite $\text{Tr}(\mathbf{J}^{-1})$ is a strictly convex function and the constraint (P2-C3) itself defines a convex set. However, the constraint involves the nonlinear matrix inverse; to make it directly processable by an interior-point SDP solver, this subsection losslessly converts it into an LMI via the Schur complement.

Introduce an auxiliary Hermitian matrix variable $\mathbf{M}_p \in \mathbb{H}^{N_\theta \times N_\theta}$ (where N_θ is the dimension of the target parameter to be estimated). The original constraint can be split equivalently as

$$\mathbf{M}_p - (\mathbf{J}_p^{\text{data}})^{-1} \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (4.5a)$$

$$\text{Tr}(\mathbf{M}_p) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}. \quad (4.5b)$$

Since $\mathbf{J}_p^{\text{data}} \succ \mathbf{0}$, by the Schur complement lemma, (4.5a) is equivalent to

$$\begin{bmatrix} \mathbf{M}_p & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_p^{\text{data}} \end{bmatrix} \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}. \quad (4.6)$$

Therefore, the PCRB constraint (P2-C3) is converted completely and without approximation into the linear matrix inequalities in $(\{\mathbf{S}_p\}, \{\mathbf{M}_p\})$:

$$\begin{bmatrix} \mathbf{M}_p & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_p^{\text{data}} \end{bmatrix} \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (4.7a)$$

$$\text{Tr}(\mathbf{M}_p) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}. \quad (4.7b)$$

This transformation introduces no relaxation gap and avoids outer SCA iterations.

E. AP–Target Soft Assignment: Continuous Relaxation with DC Penalty

This subsection relaxes the original binary condition (P1-C7') to the box constraint (P2-C9) and represents binarity on that box through a DC deficiency function. A path-loss ranking can be used only as a cold-start aid; it is not an optimality claim or a replacement for binary recovery. The DC representation is

$$0 \leq b_{mp} \leq 1, \quad (\text{E.1a})$$

$$g(b_{mp}) \triangleq b_{mp} - b_{mp}^2 = 0. \quad (\text{E.1b})$$

For $b \in [0, 1]$, $g(b) = b - b^2 \geq 0$ holds and g is a concave function of b (since $g''(b) = -2 < 0$); reformulating the equality constraint (E.1b) as a penalty term added to the objective gives

$$\min \underbrace{\sum_{k=1}^K \text{Tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p)}_{\text{original objective}} + \eta_b \underbrace{\sum_{m \in \mathcal{M}, p \in \mathcal{P}} g(b_{mp})}_{=\sum_{m,p} (b_{mp} - b_{mp}^2)} \quad \text{s.t. (P2-C1) } \sim \text{(P2-C9)} \setminus \{(\text{P2-C9})\}, \quad (\text{P2-DCP})$$

where $\eta_b \geq 0$ is the penalty factor; the objective exhibits a DC structure (“convex function + concave function $\times \eta_b$ ”) and the box constraints (P2-C9) are kept in their convex form, but the term $-\eta_b b_{mp}^2$ in the objective is concave in b_{mp} , so (P2-DCP) is not a convex optimization problem and this non-convexity is handled jointly with the rank-1 constraint by the dual DC penalty SCA framework in §3. For $b_{mp} \in \{0, 1\}$ we have $g(b_{mp}) = 0$ and the penalty incurs no cost on binary solutions; for $b_{mp} \in (0, 1)$ we have $g(b_{mp}) > 0$ and the penalty grows with η_b , thus increasing η_b encourages small binary deficiency, but no finite penalty alone guarantees $b_{mp}^* \in \{0, 1\}$. Binary recovery is certified only after the candidate is fixed and physically validated. After introducing the η_b concave penalty, (P2-DCP) has exactly the same mathematical structure as the rank-1 penalty $\rho(\mathbf{W}_k) = \text{Tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k)$ introduced in §3-A—both are “convex – convex” DC forms—so the SCA framework designed in §3 can carry both penalties simultaneously: within a single SCA iteration we perform an eigendecomposition of $\mathbf{W}_k$ and a first-order Taylor expansion of g at b_{mp} , yielding a unified convex SDP subproblem in the variables $(\{\mathbf{W}_k\}, \{b_{mp}\}, \{\mathbf{S}_p\}, \{\mu_k\}, \{\mathbf{M}_p\})$. The box constraint $0 \leq b_{mp} \leq 1$ of (P2-DCP) restricts the SCA cold start; in practice we may use a *large-scale-fading heuristic* to generate the initial point—for each target p , sort APs by $\text{PL}(d_{m,p})$ in ascending order, set $b_{mp}^{(0)} = 1$ for the top- N_{req} APs and $b_{mp}^{(0)} = 0$ for the rest; this value automatically satisfies the box constraint and the equality (P2-C5'), and is at most one step away from the $\{0, 1\}$ boundary. *This heuristic*

serves only as a cold-start aid for SCA and is no longer the final solver—the final candidate can depend on the SCA initialization and candidate-recovery path; physical validation is therefore an essential final step.

F. Consolidation (P3): Final Form with Dual DC Penalties

Applying the SDR step of §-B ((S1.1)), the S-Procedure step of §-C ((S2.1) + (S3.1)), the Schur complement of §-D ((4.7)), and the continuous relaxation + DC penalty of §-E (box (P2-C9) + (P2-DCP)) to (P2), we obtain the final problem (P3):

$$\min_{\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{\mu_k\}, \{\mathbf{M}_p\}, \{b_{mp}\}} \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2) \quad (\text{P3})$$

$$\text{s.t.} \quad \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \quad (\text{P3-C1})$$

$$\text{Tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \quad (\text{P3-C2})$$

$$\begin{bmatrix} \mathbf{M}_p & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_p^{\text{data}} \end{bmatrix} \succeq \mathbf{0}, \quad \forall p \quad (\text{P3-C3})$$

$$\text{Tr}(\mathbf{M}_p) \leq \Gamma_{\text{Track},p}, \quad \forall p \quad (\text{P3-C3b})$$

$$\text{Tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m, p \quad (\text{P3-C4'a})$$

$$\text{Tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \quad (\text{P3-C4'b})$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}} \quad (\text{P3-C5'})$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \quad (\text{P3-C6})$$

$$\mathbf{S}_p \succeq \mathbf{0}, \quad \forall p \quad (\text{P3-C7})$$

$$\mu_k \geq 0, \quad \forall k \quad (\text{P3-C8})$$

$$0 \leq b_{mp} \leq 1, \quad \forall m, p. \quad (\text{P3-C9})$$

where $\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j - \sum_p \mathbf{S}_p$, $\mathbf{J}_p^{\text{data}}$ is an affine function of $\mathbf{S}_p$, and $\mathbf{E}_m$ is the antenna-block-extraction diagonal matrix. The feasible region of (P3) is the intersection of the LMI set (P3-C1), the LMI set (P3-C3), the linear equalities/inequalities (P3-C2)/(P3-C3b)/(P3-C4'a)/(P3-C4'b)/(P3-C5'), the positive-semidefinite cones (P3-C6)/(P3-C7), the scalar half-space (P3-C8), and the box constraints (P3-C9); each of these sets is convex, so the feasible

region is convex. The objective $\sum_k \text{Tr}(\mathbf{W}_k) + \sum_p \text{Tr}(\mathbf{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2)$ is linear in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and is a “linear $-\eta_b \cdot$ quadratic” DC form in $\{b_{mp}\}$, and the overall objective is DC (non-convex) in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{b_{mp}\})$, so (P3) is not a convex SDP and its non-convexity is solved by the dual DC penalty SCA framework of §3.

Theorem 4.2 (Structural Properties of (P3)). *The feasible region of (P3) is convex; the objective is a DC function of $(\{\mathbf{W}_k\}, \{b_{mp}\}, \{\mathbf{S}_p\})$; the constraint matrices are affine in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and linear in $\{b_{mp}\}$. Hence (P3) is a difference-of-convex program (DCP), and a stationary point can be obtained by the dual DC penalty SCA framework of §3.*

Proof. See Appendix A.1. □

4.3 Joint DC Penalty SCA Framework: Rank-One and Binary Recovery

The feasible region of (P3) is convex but its objective is DC, so obtaining a global optimum directly is hard. A KKT stationary point is a candidate for recovery, not by itself a certificate of a binary rank-one physical design. This section designs a dual DC penalty SCA framework: the concave term $-\eta_b b_{mp}^2$ in the (P3) objective and the concave term $-\eta_{\text{rank}} \lambda_{\max}(\mathbf{W}_k)$ (corresponding to the missing rank-1 property) are linearized simultaneously within the same SCA loop, and each iteration solves a standard convex SDP. This section also presents Gaussian randomization / leading-eigenvector extraction as a baseline post-processing.

A. Unified Characterization of the Dual DC Penalties

For any positive-semidefinite matrix $\mathbf{W}_k \succeq \mathbf{0}$, it has rank one if and only if its trace equals its largest eigenvalue:

$$\text{rank}(\mathbf{W}_k) = 1 \iff \text{Tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k) = 0. \quad (4.9)$$

Since $\lambda_{\max}(\mathbf{W}_k) \leq \text{Tr}(\mathbf{W}_k)$ always holds for PSD matrices, define the rank-deficiency measure

$$\rho(\mathbf{W}_k) \triangleq \text{Tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k) \geq 0. \quad (4.10)$$

For $b \in [0, 1]$, define the binarity-deficiency measure

$$g(b) \triangleq b - b^2 = b(1 - b) \geq 0, \quad g''(b) = -2 < 0 \text{ (strictly concave)}. \quad (4.11)$$

$\rho(\mathbf{W}_k)$ and $g(b_{mp})$ are mathematically isomorphic—both are “convex – convex” DC forms, vanish on the desired constraint set, and are positive when the constraint is violated; this isomorphism is the mathematical foundation that allows the two penalties to share a single SCA framework. Adding the rank-1 penalty $\eta_{\text{rank}} \sum_k \rho(\mathbf{W}_k)$ to the (P3) objective gives

$$\min \underbrace{\sum_k \text{Tr}(\mathbf{W}_k) + \sum_p \text{Tr}(\mathbf{S}_p)}_{\text{original objective}} + \underbrace{\eta_{\text{rank}} \sum_k \rho(\mathbf{W}_k)}_{\text{rank-1 penalty (DC)}} + \underbrace{\eta_b \sum_{m,p} g(b_{mp})}_{\text{binarity penalty (DC)}} \quad \text{s.t. (P3-C1) } \sim \text{(P3-C9).} \quad (4.12)$$

All constraints of (P3-DCP) are convex (inherited from the convex feasible region of (P3)), and the only non-convexity comes from the two DC penalty terms in the objective.

B. SCA Linearization: Simultaneous First-Order Taylor Expansion of Both DC Terms

At iteration t , let the current iterate be $(\{\mathbf{W}_k^{(t)}\}, \{b_{mp}^{(t)}\})$. Perform the eigendecomposition

$$\mathbf{W}_k^{(t)} = \sum_{i=1}^{r_k} \lambda_{k,i} \mathbf{u}_{k,i} \mathbf{u}_{k,i}^H, \quad \lambda_{k,1} \geq \lambda_{k,2} \geq \dots \geq \lambda_{k,r_k} > 0, \quad (4.13)$$

where the leading eigenvector is $\mathbf{u}_{\max,k}^{(t)} \triangleq \mathbf{u}_{k,1}$. By the convexity of $\lambda_{\max}(\mathbf{W}_k)$, its first-order Taylor expansion provides a global lower bound:

$$\lambda_{\max}(\mathbf{W}_k) \geq \lambda_{\max}(\mathbf{W}_k^{(t)}) + \text{Tr} \left(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H (\mathbf{W}_k - \mathbf{W}_k^{(t)}) \right) = \text{Tr} \left(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k \right). \quad (4.14)$$

Substituting this into the rank-1 penalty of (P3-DCP) yields a global upper bound (the concave term is replaced by its upper bound, the subproblem minimizes, and the true objective is a lower bound of the subproblem optimum):

$$\rho(\mathbf{W}_k) = \text{Tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k) \leq \text{Tr}(\mathbf{W}_k) - \text{Tr} \left(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k \right) \triangleq \hat{\rho}^{(t)}(\mathbf{W}_k). \quad (4.15)$$

The first-order Taylor expansion of the concave $g(b) = b - b^2$ at $b^{(t)}$ gives the global overestimator

$$g(b) \leq g(b^{(t)}) + g'(b^{(t)})(b - b^{(t)}) = b^{(t)} - (b^{(t)})^2 + (1 - 2b^{(t)})(b - b^{(t)}), \quad (4.16)$$

and after removing the constant term $b^{(t)} - (b^{(t)})^2 - (1 - 2b^{(t)})b^{(t)} = (b^{(t)})^2$ (which does not affect minimization), the linearized penalty is

$$g(b) \leq (1 - 2b^{(t)})b + \text{const} \triangleq \hat{g}^{(t)}(b). \quad (4.17)$$

Substituting (4.15) and (4.17) into (P3-DCP) gives the convex SDP subproblem at iteration t :

$$\begin{aligned} \min_{\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{\mu_k\}, \{\mathbf{M}_p\}, \{b_{mp}\}} \quad & \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p) + \eta_{\text{rank}} \sum_{k=1}^K \left[\text{Tr}(\mathbf{W}_k) - \text{Tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k) \right] \\ & + \eta_b \sum_{m,p} (1 - 2b_{mp}^{(t)}) b_{mp} \end{aligned}$$

s.t. (P3-C1) $\sim$ (P3-C9).

(4.18)

The objective of (4.18) is linear in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{b_{mp}\})$ (the η_{rank} term combines $\text{Tr}(\mathbf{W}_k)$ with $-\text{Tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k)$ to form $\text{Tr}((\mathbf{I} - \mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H) \mathbf{W}_k)$, and the η_b term is linear in b_{mp}), and the constraints (P3-C1)–(P3-C9) are all LMI / linear / positive-semidefinite / box, so (P3-SCA- t) is a standard convex SDP that can be efficiently solved by interior-point solvers such as MOSEK/SDPT3. Update the iterate $(\{\mathbf{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\mathbf{S}_p^{(t+1)}\}, \{\mu_k^{(t+1)}\}, \{\mathbf{M}_p^{(t+1)}\})$ and repeat.

Theorem 4.3 (Dual DC SCA Limit-Point Property). *Assume fixed penalty factors and feasible SCA subproblems. The majorization–minimization updates produce a non-increasing sequence of penalized objective values and hence a convergent objective sequence. Finite penalty factors alone do not guarantee exact rank-one or binary recovery. If a limit point satisfies $\rho(\mathbf{W}_k^*) = 0$ for every k and $g(b_{mp}^*) = 0$ for every (m, p) , then the recovered variables satisfy the lifted constraints corresponding to (P1). Direct physical validation of the extracted beamformers and binary associations is still required to certify (P1-C1)–(P1-C7').*

Proof. See Appendix A.2. □

C. Baseline: Gaussian Randomization / Eigen-Extraction

If one pursues the fastest single-SDP run, the master problem (P3) with $\eta_{\text{rank}} = \eta_b = 0$ can be solved directly and the rank-one and binary solutions are recovered by post-processing: $\mathbf{W}_k$ is recovered as $\mathbf{w}_k^*$ via leading-eigenvector extraction and power scaling in Algorithm 2, and b_{mp} is rounded to $\{0, 1\}$ by a threshold of 0.5. It must be emphasized that this kind of post-processing breaks the strict robust-SINR guarantee of the S-Procedure and the exact satisfaction of the box constraints, and additional power scaling may be required in practice to restore feasibility.

Algorithm 1 Dual DC Penalty SCA Iteration (Joint Rank-One and Binary Recovery)

Require: Initial feasible point $\{\mathbf{W}_k^{(0)}, b_{mp}^{(0)}\}$ (generated by the heuristic of §4.2-E), penalty factors $\eta_{\text{rank}}, \eta_b > 0$, tolerance $\varepsilon > 0$, maximum number of iterations T_{max} .

Ensure: Rank-one (or approximately rank-one) beamforming covariance $\{\mathbf{W}_k^*\}$, binary AP-target assignment $\{b_{mp}^* \in \{0, 1\}\}$, sensing covariances $\{\mathbf{S}_p^*\}$.

- 1: Solve the unpenalized ($\eta_{\text{rank}} = \eta_b = 0$) problem (P3) to obtain an initial SDR solution; this solution may have $b_{mp} \notin \{0, 1\}$ when $\eta_b = 0$, but it is feasible as an SCA starting point. Report infeasible if no such solution exists.
 - 2: **for** $t = 0, 1, \dots, T_{\text{max}} - 1$ **do**
 - 3: **for** each $k \in \mathcal{K}$ **do**
 - 4: Eigendecompose $\mathbf{W}_k^{(t)}$ to obtain the leading eigenvector $\mathbf{u}_{\text{max},k}^{(t)}$.
 - 5: **end for**
 - 6: **for** each (m, p) **do**
 - 7: Compute the g' coefficient $c_{mp}^{(t)} = 1 - 2b_{mp}^{(t)}$.
 - 8: **end for**
 - 9: Solve the convex SDP (4.18) to obtain $(\{\mathbf{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\mathbf{S}_p^{(t+1)}\}, \{\mu_k^{(t+1)}\}, \{\mathbf{M}_p^{(t+1)}\})$.
 - 10: Compute $\delta_{\text{rank}}^{(t)} = \sum_k \rho(\mathbf{W}_k^{(t+1)})$ and $\delta_b^{(t)} = \sum_{m,p} g(b_{mp}^{(t+1)})$.
 - 11: **if** $\delta_{\text{rank}}^{(t)} \leq \varepsilon$ and $\delta_b^{(t)} \leq \varepsilon$ and the change in objective $|P^{(t+1)} - P^{(t)}| \leq \varepsilon$ **then**
 - 12: **break**.
 - 13: **end if**
 - 14: **end for**
 - 15: For each k , set $\mathbf{w}_k^* = \sqrt{\lambda_{\text{max}}(\mathbf{W}_k^{(t+1)})} \mathbf{u}_{\text{max},k}^{(t+1)}$.
 - 16: For each (m, p) , round $b_{mp}^* = \mathbb{I}[b_{mp}^{(t+1)} \geq 0.5]$ (if $\delta_b^{(t)}$ is sufficiently small, the rounding incurs no cost).
 - 17: **return** $\{\mathbf{w}_k^*, b_{mp}^*, \{\mathbf{S}_p^*\}\}$.
-

Algorithm 2 Rank-One Beamforming Extraction (Baseline Post-Processing)

Require: Converged solution $\{\mathbf{W}_k^*, \{\mathbf{S}_p^*\}, b_{mp}^*\}$.

Ensure: Rank-one beamforming vectors $\{\mathbf{w}_k^*\}$.

- 1: **for** each $k \in \mathcal{K}$ **do**
 - 2: Compute the eigendecomposition $\mathbf{W}_k^* = \mathbf{U}_k \mathbf{\Lambda}_k \mathbf{U}_k^H$.
 - 3: Extract the leading eigenvector: $\tilde{\mathbf{w}}_k = \sqrt{\lambda_{\max}(\mathbf{\Lambda}_k)} \cdot \mathbf{u}_k^{\max}$.
 - 4: Scale to satisfy the per-AP power: $\mathbf{w}_k^* = \eta_k \tilde{\mathbf{w}}_k$, where η_k is chosen so that $\text{Tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}$ with $\mathbf{R}_X = \sum_k \mathbf{w}_k^* (\mathbf{w}_k^*)^H + \sum_p \mathbf{S}_p^*$ for every m .
 - 5: **end for**
 - 6: **if** $\{\mathbf{w}_k^*, b_{mp}^*\}$ satisfies all of (P1-C1)–(P1-C7') **then**
 - 7: **return** $\{\mathbf{w}_k^*, b_{mp}^*\}$.
 - 8: **else**
 - 9: Apply a power-scaling fallback: $\mathbf{w}_k^* \leftarrow \alpha \mathbf{w}_k^*$, with $\alpha \in (0, 1]$ chosen to restore feasibility.
 - 10: **return** $\{\mathbf{w}_k^*, b_{mp}^*\}$.
 - 11: **end if**
-

4.4 Link to Physical Evaluation

The convexification chain represents the robust SINR and PCRB conditions by LMIs in the lifted variables, while the dual DC procedure drives the rank and binary deficiencies toward zero. Neither a finite penalty nor a stationary point alone proves exact recovery or global optimality. The final step is therefore candidate-specific: recover or re-solve the binary topology, extract the beamformers, and directly validate (P1-C1)–(P1-C7'). Chapter 5 uses this distinction to compare methods by physical feasibility first and conditional performance second.

Chapter 5

Numerical Evaluation and Results

Analysis

This chapter tests the three links in the thesis contribution chain: whether topology-aware recovery improves physically realizable sensing designs, whether the robust formulation protects communication QoS under CSI uncertainty, and whether the complete workflow remains useful after recovery and at a larger network dimension. Each result is interpreted with its feasibility evidence, not only with a conditional performance average.

5.1 Evaluation Protocol, Metrics, and Baselines

The experiments use the result sets stored with the paper figures. A run is counted as physically feasible only after binary association recovery, rank-one beamformer extraction or fixed-topology re-solution, and direct validation of the recovered design. The validation checks the robust communication SINR, sensing SINR, matrix PCRB, per-AP transmit-power limit, and association cardinality. Consequently, an SDR solution or a continuous association vector is not reported as a feasible physical design by itself.

All compared methods use common random scenarios within a sweep. Feasibility rates are therefore unconditional statistics over those scenarios. Power, runtime, rate, sensing-SINR, and PCRB summaries marked “conditional” are computed only over the recovered designs that pass physical validation. The baselines separate different design choices: FIM-greedy and nearest-AP methods fix topology according to sensing geometry, random topology removes geometry awareness, and relaxation or single-penalty variants isolate the recovery mechanism. This pro-

protocol ensures that a low conditional power is never read as an advantage when it is accompanied by frequent infeasibility.

5.2 Topology-Recovery Advantage in Ordinary and Stress Geometries

Figure 5.1 compares topology-recovery strategies over 30 common seeds for a six-AP configuration. The proposed full recovery and the FIM-greedy topology are physically feasible in all displayed cluster-size settings. In contrast, the nearest-AP baseline is infeasible in 20% of cases at $N_{\text{req}} = 2$, and the random-topology baseline is substantially less reliable for small and intermediate cluster sizes. The comparison shows that the recovery workflow can identify physically usable sensing support where purely local or random topology rules fail; the comparison is not evidence that relaxation alone is exact.

The cardinality equality in (P1-C5') is a resource-allocation requirement, not a monotone performance knob. Increasing N_{req} changes the authorized sensing geometry and the feasible power allocation simultaneously, so conditional power need not vary monotonically. The PCRB ratios remain close to one in feasible recovered cases, which is consistent with the tracking constraint being active at the solution.

Figure 5.2 asks whether the same topology advantage persists when the geometry is deliberately unfavorable. With colocated edge users and targets, and again with crowded targets, the proposed full recovery and FIM-greedy topology remain feasible in all shown cases. The nearest-AP topology loses feasibility in the edge-colocated setting and falls to roughly 24% in the crowded setting. When it succeeds in the crowded case, its conditional power approaches 300 mW, whereas the proposed and FIM-greedy methods remain near 50 mW. The physical mechanism is geometric: selecting the nearest APs alone can exhaust useful sensing directions or AP power under crowding. The evidence is limited to the tested pressure configurations, but it demonstrates why feasibility must accompany every conditional comparison.

5.3 Robustness–Power Trade-Off under CSI Uncertainty

Figure 5.3 compares the robust design with a nominal-CSI baseline as the normalized CSI-uncertainty radius increases. Both methods are certified feasible at their respective design

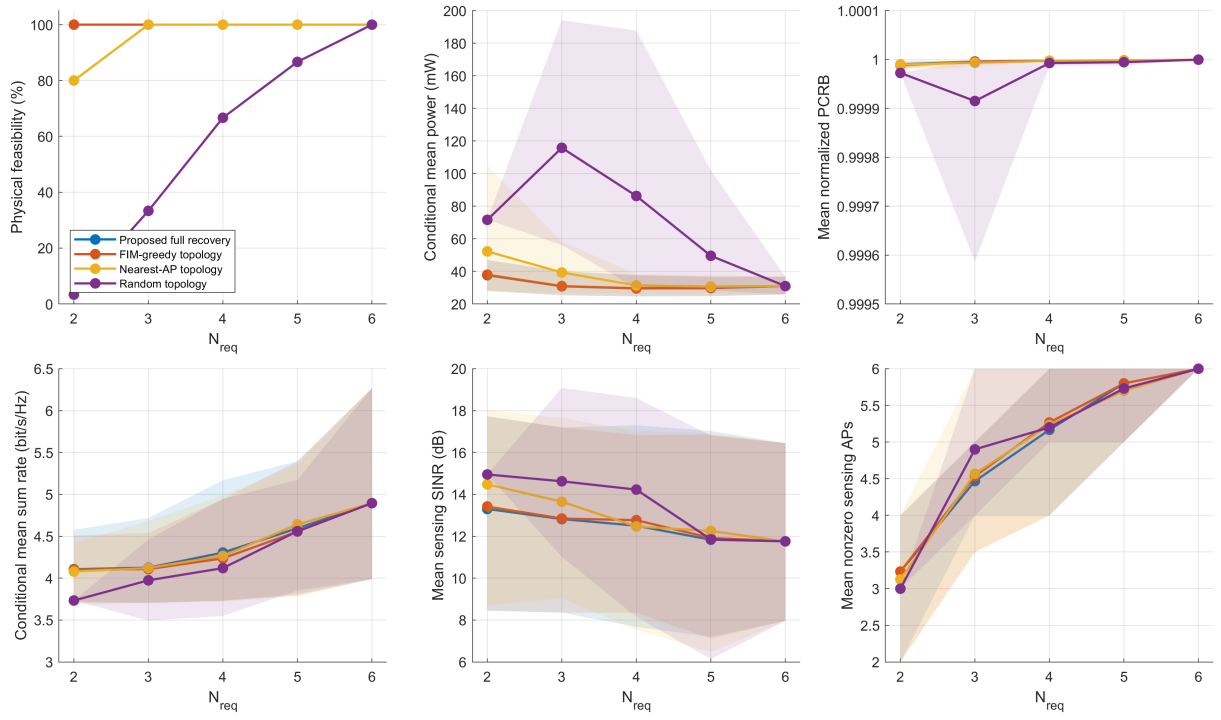


Figure 5.1: Topology-method comparison versus required sensing APs per target. Feasibility is measured over all common scenarios; power, rate, sensing-SINR, and PCRB summaries are conditional on physically feasible recovered designs.

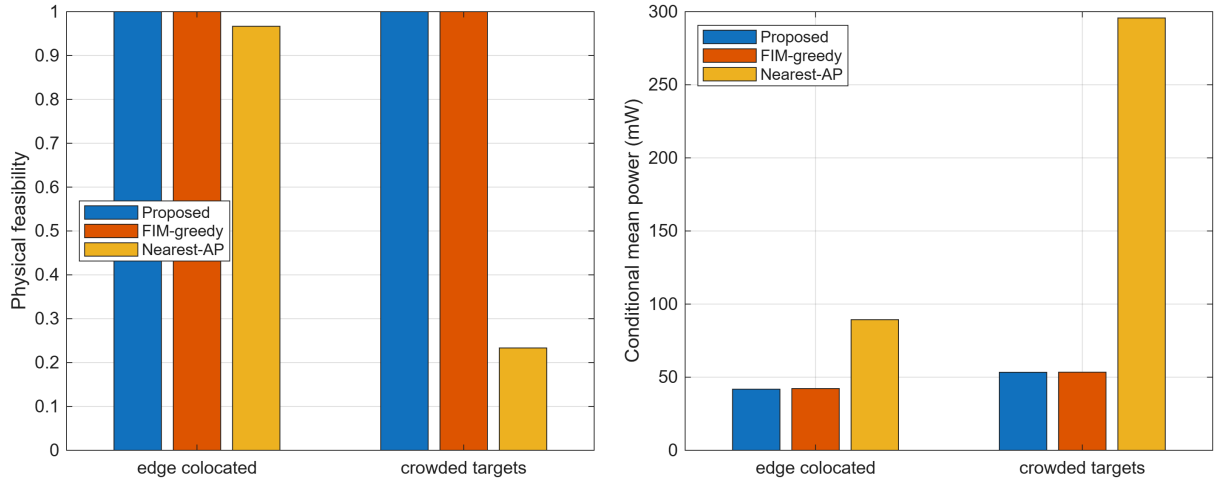


Figure 5.2: Physical feasibility and conditional power in two pressure geometries. Conditional-power bars exclude infeasible recovered runs and must be interpreted together with the associated feasibility rate.

points. When tested against perturbations within the uncertainty ball, the robust design has zero observed outage throughout the displayed sweep, whereas the nominal-CSI design has approximately 90% outage for nonzero uncertainty radii. The robust protection raises transmit power moderately, from about 28.7 mW at zero uncertainty to about 30 mW at the largest displayed radius.

This is the intended robustness–power trade-off: the additional power is associated with markedly improved robustness under the tested perturbations. It should not be interpreted as a guarantee outside the specified norm-bounded uncertainty model, nor as a claim that the nominal design is infeasible at the channel estimate itself.

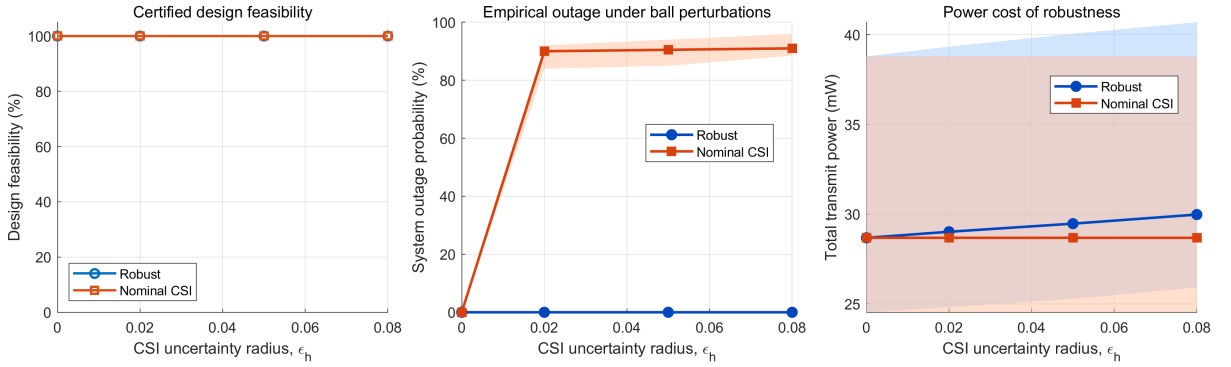


Figure 5.3: Robustness comparison under increasing CSI-uncertainty radius. The middle panel is an empirical perturbation test and the right panel reports the corresponding transmit-power cost.

5.4 Recovery Mechanism and DC-SCA Stabilization

The preceding comparisons concern recovered physical designs. Figure 5.4 therefore isolates the mechanism that makes recovery reliable. Across 25 common scenarios, all four variants have the same observed physical-feasibility rate of 92%, so the ablation does not show that one penalty alone enlarges the feasible set in this test. Its informative result is recovery quality: relaxation and rank-only variants retain a binary distance on the order of 10^{-1} , whereas binary-only and dual-DC variants reduce the median binary distance below the 10^{-4} reference. The dual-DC variant also maintains a very small median rank residual. Thus both penalties are needed when the final design must be binary and rank one simultaneously.

Figure 5.5 provides the iteration-level complement to the ablation on the same 25 seeds. The median binary distance falls from approximately 0.43 at initialization to below the recovery

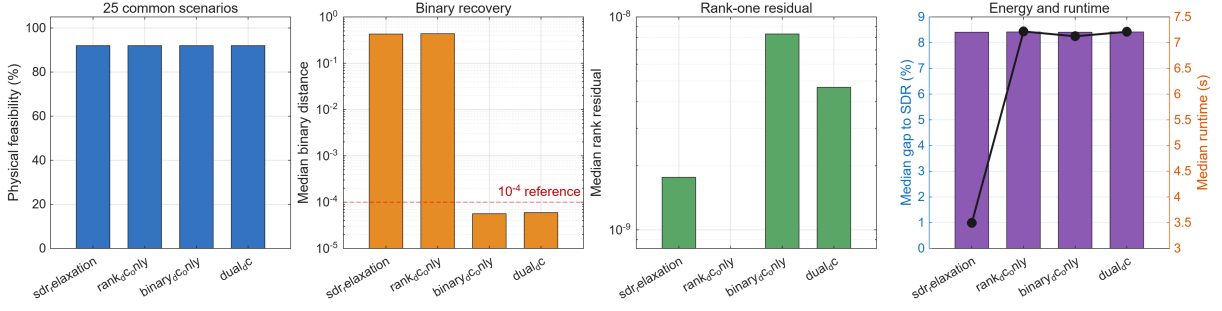


Figure 5.4: Recovery ablation over 25 common scenarios. Physical feasibility is evaluated after recovery; binary distance and rank residual diagnose whether a relaxed solution is suitable for physical recovery.

threshold after the first DC-SCA iteration, and the recovery-ready fraction remains at 100% over the displayed iterations. The median rank residual stays at the 10^{-9} scale. The continuous-power trace is a continuation diagnostic only; it is not used to assert monotone descent of the original non-convex objective.

5.5 Larger-Scale Validation and Computational Cost

Figure 5.6 examines a larger configuration with $M = 12$, $K = 6$, $P = 3$, and $N_{\text{req}} = 3$. Over eight common seeds, the proposed full recovery is physically feasible in all cases, whereas the FIM-greedy and nearest-AP baselines achieve 75%, and the random topology is feasible in only 12.5% of cases. The proposed conditional energy cost is close to the FIM-greedy cost and substantially below the nearest-AP and random topologies. Its wall-clock cost is higher than that of fixed-topology baselines because the workflow performs candidate generation, fixed-topology re-solves, and physical validation.

These observations provide scalability evidence at the tested dimensions, not an asymptotic complexity claim. The smaller number of seeds also means that larger independent sweeps and confidence intervals are required before drawing stronger quantitative conclusions.

5.6 Summary of Findings

The numerical evidence supports three bounded conclusions. First, topology-aware recovery improves physical feasibility relative to nearest and random topology rules in the tested ordinary and pressure geometries. Second, the robust formulation exchanges a modest power

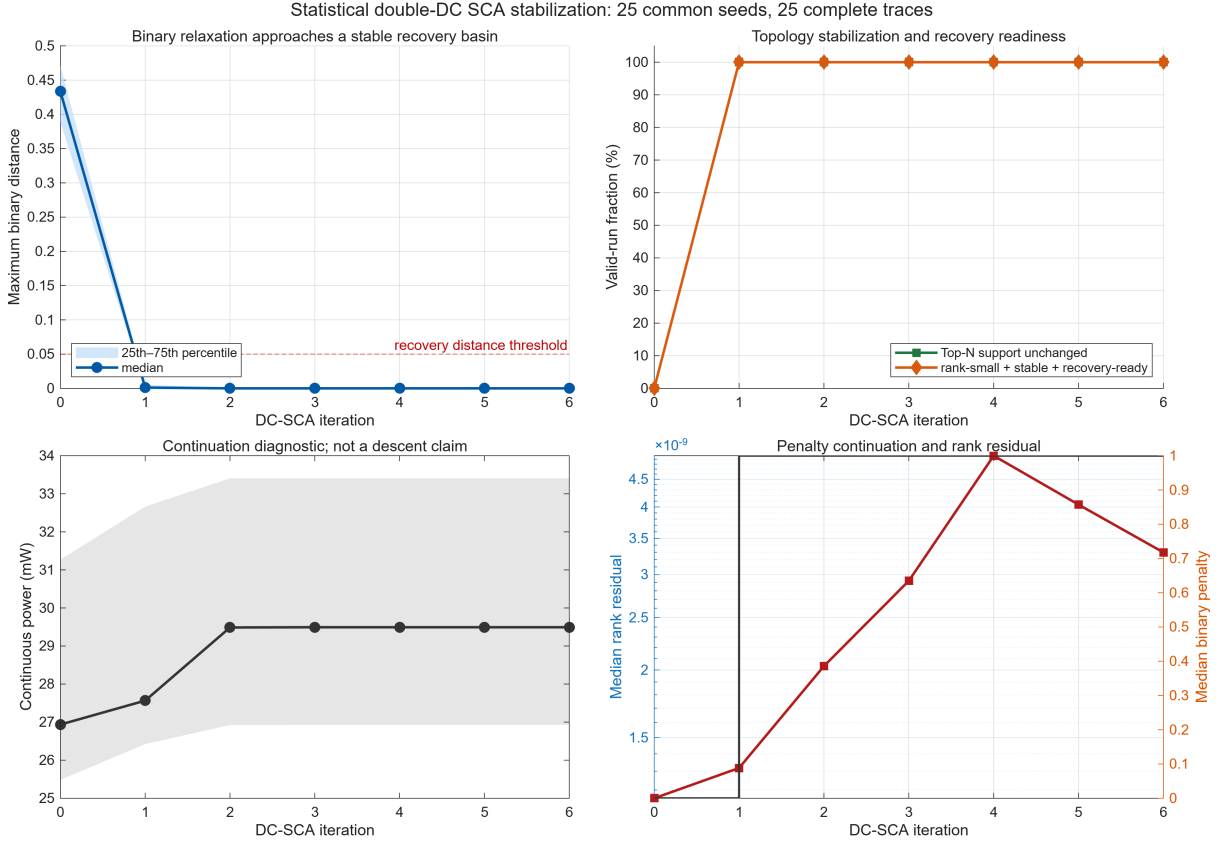


Figure 5.5: Statistical dual-DC SCA stabilization over 25 complete traces. The recovery-ready statistic combines a small binary distance, a small rank residual, and stable topology support.

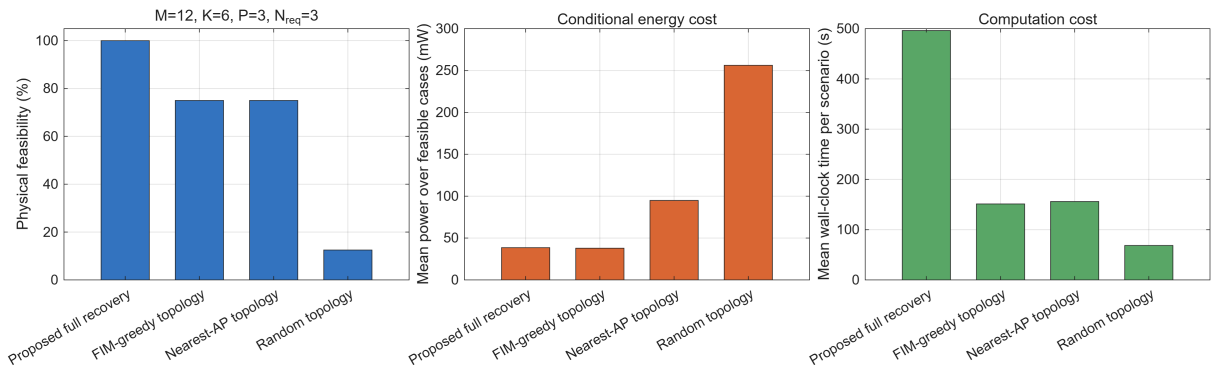


Figure 5.6: Scalability validation for $M = 12$, $K = 6$, $P = 3$, and $N_{\text{req}} = 3$ over eight common seeds. Power and runtime are conditional summaries for physically feasible recovered designs.

increase for substantially improved outage behaviour within the stated CSI-uncertainty model. Third, the dual DC penalties provide recovery-quality evidence, while the larger-scale experiment indicates that the validated workflow remains usable beyond the smallest configuration. These conclusions remain empirical: they do not prove global optimality, universal relaxation tightness, or asymptotic scalability.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis investigated how a cell-free ISAC transmitter can minimize total transmit power while providing robust communication QoS and target-specific sensing service with a physically implementable AP–target topology. The resulting design keeps communication beam-forming globally cooperative, while binary association variables authorize only the dedicated sensing covariance of each target. This separation aligns the sensing topology, the target-specific FIM/PCRB requirement, and the dedicated sensing power directly controlled by the optimization.

The formulated mixed-integer problem jointly accounts for worst-case communication SINR, sensing SINR, matrix PCRB, dedicated-sensing authorization, and total AP-power limits. Covariance lifting, the S-Procedure, and a Schur-complement representation yield solver-ready robust and tracking constraints. A dual DC-penalty SCA procedure then reduces both rank-one and binary deficiencies, and direct validation distinguishes a recovered physical design from a relaxed optimization point.

The numerical results support three bounded conclusions. First, topology-aware recovery improves physical feasibility relative to nearest-AP and random topology rules in the tested ordinary and pressure geometries. Second, robust design incurs a modest transmit-power increase while markedly reducing observed outage within the specified CSI-uncertainty model. Third, the dual-DC workflow provides recovery-quality evidence through small rank and binary residuals before physical validation. The larger-dimensional experiment also demonstrates usable recovery behaviour at the tested scale, at a higher computational cost than fixed-topology

baselines. These are empirical, configuration-specific findings; they do not establish global optimality, universal relaxation tightness, or asymptotic scalability.

6.2 Future Work

The immediate priority is broader, reproducible validation. Future experiments should report feasibility, recovered physical transmit power, worst-case communication SINR, sensing SINR, PCRB, per-AP power, and association-cardinality residuals for every sweep, while preserving the distinction between relaxed and recovered solutions. Larger independent seed sets, confidence intervals, and systematic PCRB, cluster-size, and CSI-error sweeps are needed to quantify the robustness–sensing–power trade-off more reliably.

The model can also be extended beyond its single-interval assumptions through dynamic target-state prediction across scheduling intervals, uncertainty in sensing channels and target parameters, fronthaul or processing costs for sensing clusters, and distributed implementations for larger cell-free networks. Each extension should retain direct validation of the recovered physical variables so that additional modelling flexibility does not obscure the communication, sensing, tracking, and AP-power guarantees.

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Appendix A

Mathematical Proofs

A.1 Proof of Theorem 4.2

This appendix proves Theorem 4.2.

The feasible region of (P3) is the intersection of the LMI set (P3-C1), the linear equalities/inequalities (P3-C2)/(P3-C3b)/(P3-C4'a)/(P3-C4'b)/(P3-C5'), the LMI set (P3-C3), the positive-semidefinite cones (P3-C6)/(P3-C7), the scalar half-space (P3-C8), and the box constraints (P3-C9). The LMI set, the linear-inequality/equality set, the positive-semidefinite cone, the scalar half-space, and the box set are all convex; the intersection of finitely many convex sets is convex, so the feasible region of (P3) is convex.

The objective $F(\{\mathbf{W}_k\}, \{b_{mp}\}, \{\mathbf{S}_p\}) = \sum_k \text{Tr}(\mathbf{W}_k) + \sum_p \text{Tr}(\mathbf{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2)$ is linear in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and is a “linear $-\eta_b \cdot$ quadratic” DC form in $\{b_{mp}\}$. The constraint matrices are affine in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and linear in $\{b_{mp}\}$, so (P3) is a DC program.

For complexity, the total number of decision variables is $n = K \cdot (MN_t)^2 + P \cdot (MN_t)^2 + K + P \cdot N_\theta^2 + MP$ (K matrices $\mathbf{W}_k \in \mathbb{H}_+^{MN_t}$, P matrices $\mathbf{S}_p \in \mathbb{H}_+^{MN_t}$, K S-Procedure relaxation variables μ_k , P auxiliary matrices $\mathbf{M}_p \in \mathbb{H}_+^{N_\theta \times N_\theta}$, and MP scalars b_{mp}); the number of inequality constraints is $m = K + P + P + P + MP + M + P_{\text{active}} + K + P + K + 2MP = 3K + 4P + M + 3MP + P_{\text{active}}$ ((P3-C1): K ; (P3-C2): P ; (P3-C3): P ; (P3-C3b): P ; (P3-C4'a): MP ; (P3-C4'b): M ; (P3-C5'): P_{active} ; (P3-C6): K ; (P3-C7): P ; (P3-C8): K ; (P3-C9): $2MP$); the largest LMI constraint size is $\ell = \max\{MN_t + 1, 2N_\theta\}$. Each SCA subproblem (P3-SCA- t) is a standard convex SDP, solvable in polynomial time by interior-point methods.

A.2 Proof of Theorem 4.3

This appendix proves the convergence of the objective sequence of Algorithm 1 and the limiting property $b_{mp} \rightarrow \{0, 1\}$.

Denote the objective of the t -th SCA subproblem (4.18) by

$$P^{(t)}(\mathbf{W}_k, b_{mp}) \triangleq \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p) + \eta_{\text{rank}} \sum_{k=1}^K \left[\text{Tr}(\mathbf{W}_k) - \text{Tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k) \right] + \eta_b \sum_{m,p} (1 - 2b_{mp}^{(t)})$$

Let the optimal solution of the t -th iteration be $(\{\mathbf{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\mathbf{S}_p^{(t+1)}\}, \dots)$. Define the true penalty objective as

$$\tilde{P}^{(t)} \triangleq \sum_{k=1}^K \text{Tr}(\mathbf{W}_k^{(t+1)}) + \sum_{p=1}^P \text{Tr}(\mathbf{S}_p^{(t+1)}) + \eta_{\text{rank}} \sum_{k=1}^K \rho(\mathbf{W}_k^{(t+1)}) + \eta_b \sum_{m,p} g(b_{mp}^{(t+1)}).$$

Key inequality (1). Since $\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H$ is a subgradient of $\lambda_{\max}(\mathbf{W}_k)$ at $\mathbf{W}_k^{(t)}$, by the first-order condition for convex functions,

$$\lambda_{\max}(\mathbf{W}_k^{(t+1)}) \geq \text{Tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k^{(t+1)}),$$

equivalently $\rho(\mathbf{W}_k^{(t+1)}) \leq \text{Tr}(\mathbf{W}_k^{(t+1)}) - \text{Tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k^{(t+1)})$. Likewise, the first-order Taylor expansion of the concave g at $b_{mp}^{(t)}$ gives $g(b_{mp}^{(t+1)}) \leq (1 - 2b_{mp}^{(t)})b_{mp}^{(t+1)} + (b_{mp}^{(t)})^2$ (where $(b_{mp}^{(t)})^2$ is a constant). Combining both inequalities yields

$$\tilde{P}^{(t)} \leq P^{(t)}(\mathbf{W}_k^{(t+1)}, b_{mp}^{(t+1)}) + \eta_b \sum_{m,p} (b_{mp}^{(t)})^2.$$

Key inequality (2). Taking the iterate $(\{\mathbf{W}_k^{(t)}\}, \{b_{mp}^{(t)}\})$ as a feasible point of the t -th subproblem (constraints (P3-C1)–(P3-C9) are still satisfied at the iterate, since (P3-C1)–(P3-C9) are convex in $(\mathbf{W}_k, \{\mathbf{S}_p\}, b_{mp})$), its subproblem objective value is

$$\begin{aligned} P^{(t)}(\mathbf{W}_k^{(t)}, b_{mp}^{(t)}) &= \sum_k \text{Tr}(\mathbf{W}_k^{(t)}) + \sum_p \text{Tr}(\mathbf{S}_p^{(t)}) \\ &\quad + \eta_{\text{rank}} \sum_k \left[\text{Tr}(\mathbf{W}_k^{(t)}) - \text{Tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k^{(t)}) \right] \\ &\quad + \eta_b \sum_{m,p} (1 - 2b_{mp}^{(t)})b_{mp}^{(t)} \\ &= \sum_k \text{Tr}(\mathbf{W}_k^{(t)}) + \sum_p \text{Tr}(\mathbf{S}_p^{(t)}) + \eta_{\text{rank}} \sum_k \rho(\mathbf{W}_k^{(t)}) + \eta_b \sum_{m,p} (b_{mp}^{(t)} - 2(b_{mp}^{(t)})^2) \\ &= \tilde{P}^{(t-1)} - \eta_b \sum_{m,p} (b_{mp}^{(t)})^2, \end{aligned}$$

where the third line uses $\text{Tr}(\mathbf{u}_{\max,k}^{(t)}(\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k^{(t)}) = \lambda_{\max}(\mathbf{W}_k^{(t)})$ (by the leading-eigenvector definition) and the fourth line uses $g(b_{mp}^{(t)}) = b_{mp}^{(t)} - (b_{mp}^{(t)})^2$ to compare with the definition of $\tilde{P}^{(t-1)}$ (the difference is exactly $-\eta_b \sum (b_{mp}^{(t)})^2$). Since $(\{\mathbf{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\})$ is optimal for the subproblem,

$$P^{(t)}(\mathbf{W}_k^{(t+1)}, b_{mp}^{(t+1)}) \leq P^{(t)}(\mathbf{W}_k^{(t)}, b_{mp}^{(t)}) = \tilde{P}^{(t-1)} - \eta_b \sum_{m,p} (b_{mp}^{(t)})^2.$$

Monotone non-increase. Combining the two inequalities gives

$$\tilde{P}^{(t)} \leq (\tilde{P}^{(t-1)} - \eta_b \sum_{m,p} (b_{mp}^{(t)})^2) + \eta_b \sum_{m,p} (b_{mp}^{(t)})^2 = \tilde{P}^{(t-1)},$$

i.e. the true objective sequence $\{\tilde{P}^{(t)}\}$ is monotonically non-increasing.

Convergence. The true objective $\tilde{P}^{(t)}$ is monotonically non-increasing and physically bounded below by 0 (power terms and penalty terms are non-negative); by the monotone convergence theorem $\{\tilde{P}^{(t)}\}$ converges.

Recovery condition. The nonnegative deficiencies $g(b)$ and $\rho(\mathbf{W})$ quantify departure from binary and rank-one structure. Increasing the associated penalties encourages smaller deficiencies, but a finite penalty does not by itself prove that either deficiency converges to zero. If an obtained limit point has $g(b_{mp}^*) = 0$ and $\rho(\mathbf{W}_k^*) = 0$, then $b_{mp}^* \in \{0, 1\}$ and $\text{rank}(\mathbf{W}_k^*) = 1$. These conditions, together with direct constraint validation after beamformer extraction, certify recovery for that solution.

Recovery of the original constraints. At a limit point with $(\rho, g) = (0, 0)$, the recovered $\mathbf{w}_k^* = \sqrt{\lambda_{\max}(\mathbf{W}_k^*)} \mathbf{u}_{\max,k}^*$ and $b_{mp}^* \in \{0, 1\}$ satisfy the lifted constraints through the transformation chain of (P3). The rank-one property permits beamformer extraction, but the extraction and fixed binary associations must be substituted back into (P1-C1)–(P1-C7') and directly validated. Only this final validation certifies feasibility of the recovered physical design.

Appendix 1

Risk Management Form